Effective-temperature multipoles for relativistic transport. II. Direction-dependent chemical potential as a non-redundant degree of freedom

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In relativistic kinetic theory with angular anisotropy, the distribution along each propagation direction takes the equilibrium spectral shape $\Phi_\xi(x/\Theta(\hat{e}) - \eta(\hat{e})) = [e^{x/\Theta-\eta} - \xi]^{-1}$, parameterised by a direction-dependent effective temperature $\Theta(\hat{e})$ and degeneracy parameter $\eta(\hat{e})$, with $\xi = 0$ (Maxwell–Boltzmann), -1 (Fermi–Dirac), or $+1$ (Bose–Einstein). We ask whether a scalar (direction-independent) chemical potential is sufficient, or whether a direction-dependent $\eta(\hat{e})$ carries genuinely independent nonequilibrium information. We construct the minimal extension ($L_\Theta = 2$, $L_\eta = 1$) of the effective-temperature multipole manifold, adding a single dipolar chemical-potential coefficient A_η to the temperature monopole, dipole, and quadrupole $(\Theta_0, A_\Theta, Q_\Theta)$ and a scalar degeneracy η_0. This produces a five-parameter family mapped to five independent observables—three energy-density moments and two number-density moments. We establish four results for all three quantum statistics. (i) The manifold is well-posed in the coordinate $x = E/(k_{\mathrm{B}}T_0)$, eliminating the direction-dependent domain pathology of the original energy variable (Theorem 1). (ii) Numerical evaluation over all tested configurations finds the 5×5 moment-map Jacobian to be full-rank, with rank stability verified across five orders of magnitude of finite-difference step size. (iii) No scalar chemical potential, combined with arbitrary temperature multipole adjustment, can reproduce the joint energy–number moment structure generated by a chemical-potential dipole (Theorem 5). (iv) The observable signature of A_η is a shift in the number-to-energy flux fraction ratio $f_{\mathcal{N}}/f_{\mathcal{E}}$, which is exactly invariant under scalar η_0 for all ξ (Proposition 19) and responds at first order in A_η with a statistics-dependent coefficient $C_\xi(\eta_0)$ (Proposition 20). For Maxwell–Boltzmann statistics, C_ξ is η_0-independent; for Fermi–Dirac it decreases with degeneracy; for Bose–Einstein it decreases from the condensation edge toward the bulk and converges to the classical value. The paper establishes structural non-redundancy and identifiability of the chemical-potential dipole across all three quantum statistics, not a transport-performance advantage.

Scope note. Theorems 1–5 and Theorem 14 are proved analytically under stated hypotheses. Proposition 10(c) is proved via a polylogarithm Cauchy–Schwarz inequality; Proposition 10(b) and Proposition 20(b,c) are upgraded via a spectral log-concavity lemma (Lemma 9). Full monotonicity of κ (FD) and C_ξ (FD/BE) is confirmed numerically (Remarks 11 and 21). Proposition 4 is verified computationally over tested configurations.

I. INTRODUCTION

In kinetic systems with a conserved charge—neutrino transport in dense matter, photon transport with stimulated emission, or any classical radiation with a conserved species—the angular distribution of particle number can develop a structure distinct from that of the energy distribution. A transport closure that locks the number and energy sectors together by parameterising both through a single field $\Theta(\hat{e})$ cannot represent this decoupling and will misattribute number-flux anisotropy to spectral distortion. Paper I [1] introduced the effective-temperature (Teff) representation for the intensity sector; the present paper asks whether the chemical potential carries genuinely independent nonequilibrium information or is an absorbable reparameterisation.

We consider an angular-multipole representation of anisotropic distributions in relativistic kinetic theory, in which the occupation number along each propagation direction $\hat{e}$ takes the equilibrium spectral shape with a direction-dependent effective temperature $\Theta(\hat{e})$ rescaling the energy argument. In this representation—an extension of the effective-temperature (Teff) construction introduced in [1]—the chemical potential μ enters through a shifted energy variable $y = (E - \mu)/(k_{\mathrm{B}}T_0)$, and its value is the same along every direction. For applications where particle number is conserved—any kinetic system with a conserved charge, including neutrino transport in dense matter (Fermi–Dirac), photon transport in the presence of stimulated emission (Bose–Einstein), and classical radiation with a conserved species count (Maxwell–Boltzmann)—this restriction raises a structural question.

Number conservation in the presence of angular anisotropy requires the local particle-number flux to be a dynamical quantity. In a moment-based description, this flux is determined by the dipole moment of the number-density kernel $\rho_N(\hat{e})$. If the chemical potential is a scalar, then $\rho_N(\hat{e})$ is determined entirely by $\Theta(\hat{e})$: the energy and number sectors are locked. A separate angular structure in the chemical potential would decouple these sectors. The question is whether this decoupling is genuine—whether a direction-dependent $\eta(\hat{e})$ carries in-

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formation that cannot be absorbed into the temperature multipoles—or whether it is a redundant reparameterisation.

Anisotropic hydrodynamics in the Romatschke–Strickland tradition [2] addresses number conservation by introducing a scalar fugacity alongside the momentum-space anisotropy parameter [3]. Within the scalar-degeneracy subfamily of the present manifold, we find that the number-to-energy flux fraction ratio $f_\mathcal{N}/f_\mathcal{E}$ is exactly invariant under changes in η_0 for all three quantum statistics (Proposition 19): a direction-independent chemical potential cannot adjust this ratio. A direction-dependent degeneracy parameter is the minimal extension that decouples the number and energy sectors.

We construct the ($L_\Theta = 2$, $L_\eta = 1$) extension and establish, for Maxwell–Boltzmann ($\xi = 0$), Fermi–Dirac ($\xi = -1$), and Bose–Einstein ($\xi = +1$):

(i) Well-posedness of the manifold in $x = E/(k_\mathrm{B}T_0)$ coordinates, eliminating the domain pathology that arises when μ varies with direction (Theorem 1, Sec. II).

(ii) Full-rank 5×5 Jacobian of the moment map at all tested configurations, stable across five orders of magnitude of finite-difference step size (Sec. II D).

(iii) Genuine insufficiency of scalar η: no scalar chemical potential combined with adjusted temperature multipoles can match the moments of a dipole η, on the pure-dipole temperature subfamily (Theorem 5, Sec. II F).

(iv) Observable signature: the flux fraction ratio $f_\mathcal{N}/f_\mathcal{E}$ is exactly invariant under scalar η_0 (Proposition 19) and shifts at first order in A_η with a statistics-dependent coefficient $C_\xi(\eta_0)$ (Proposition 20, Sec. III F).

Five known-limit checks are reported in Sec. IV. The paper establishes structural non-redundancy and identifiability, not a transport-performance advantage. We do not couple the manifold to a transport solver, present astrophysical benchmarks, or prove the entropy-projection inequality for the extended system; these are deferred to Paper III [4] and future work.

II. THE EXTENDED EFFECTIVE-TEMPERATURE MANIFOLD

A. Coordinate choice and domain

In the original Teff construction [1], the distribution along each direction $\hat{e}$ is

$$f_\Theta = \Phi_\xi\left(\frac{y}{\Theta(\hat{e})}\right), \qquad y = \frac{E - \mu}{k_\mathrm{B}T_0}, \tag{1}$$

with $\Theta(\hat{e}) > 0$. When μ is a prescribed scalar, $y \in (0, \infty)$ for all directions and the Laguerre energy-mode basis of [1] applies. If $\mu \to \mu(\hat{e})$, then $y(\hat{e}) = (E - \mu(\hat{e}))/(k_\mathrm{B}T_0)$, and rays with $\mu(\hat{e}) > E$ produce $y < 0$, outside the Laguerre domain.

We work instead in

$$x = \frac{E}{k_\mathrm{B}T_0} \in (0, \infty), \tag{2}$$

which is direction-independent. The chemical potential enters the exponential argument:

$$f(x, \hat{e}) = \Phi_\xi\left(\frac{x}{\Theta(\hat{e})} - \eta(\hat{e})\right) = \left[\exp\left(\frac{x}{\Theta} - \eta\right) - \xi\right]^{-1}, \tag{3}$$

with $\eta(\hat{e}) = \mu(\hat{e})/(k_\mathrm{B}T_0)$. The argument $z = x/\Theta - \eta$ must satisfy $z > 0$ for $\xi = +1$ (Bose–Einstein) to ensure $f > 0$; for $\xi = 0$ and $\xi = -1$ there is no such restriction.

Theorem 1 (x-coordinate well-posedness, all ξ)**.** *The two-field family* (3) *with $x = E/(k_\mathrm{B}T_0) > 0$ is well-defined for arbitrary direction-dependent $\eta(\hat{e})$, provided $\Theta(\hat{e}) > 0$ and, for BE ($\xi = +1$), $\eta(\hat{e}) \le 0$ for all $\hat{e}$. The domain $x \in (0, \infty)$ is direction-independent, eliminating the domain pathology of the shifted variable y at $\mu > 0$.*

Proof. For MB ($\Phi_0 = e^{-z}$), $f > 0$ for all real z. For FD ($\Phi_{-1} = (e^z + 1)^{-1}$), $f > 0$ for all real z. For BE ($\Phi_{+1} = (e^z - 1)^{-1}$), positivity requires $z > 0$ for all $x > 0$, which imposes $\eta(\hat{e}) \le 0$ (since $\inf_{x>0} x/\Theta = 0$). In all cases $x \in (0, \infty)$ is direction-independent. □

The energy-mode inner product is

$$\langle g, h \rangle_* = \int_0^\infty g(x)\, h(x)\, f(1+\xi f)\, x^3\, \mathrm{d}x, \tag{4}$$

with weight $w_* = f(1+\xi f)\, x^3 > 0$ for all $x > 0$, $\Theta > 0$, and admissible η. The Gram matrix of $\{1, x\}$ is positive definite at all tested (Θ, η) configurations (Table I). (Paper I [1] uses the Laguerre weight $w_L = y^{-1}e^{-y}$ in the variable $y = (E - \mu)/(k_\mathrm{B}T_0)$; the present weight w_* is the Fisher metric weight adapted to the x-coordinate and general ξ. The tangency biconditional of Paper I holds in both inner products; quantitative diagnostic values differ by a statistics-dependent factor.)

B. Tangent space geometry

The tangent directions at (Θ, η) are

$$\frac{\partial f}{\partial \Theta} = f(1+\xi f)\, \frac{x}{\Theta^2}, \qquad \frac{\partial f}{\partial \eta} = f(1+\xi f). \tag{5}$$

Proposition 2 (Tangent-space independence, all ξ)**.** *For any $\Theta > 0$ and admissible η, the tangent vectors $\partial_\Theta f$ and $\partial_\eta f$ of the distribution* (3) *are linearly independent in the inner product* (4)*. Equivalently, the 2×2 Gram matrix G is positive definite.*

Proof. The tangent vectors are proportional to $v_1(x) = x/\Theta^2$ and $v_2(x) = 1$ in the weighted space. The Gram determinant vanishes only if v_1 and v_2 are proportional a.e. with respect to w_*. Since x/Θ^2 is not a constant function on $(0,\infty)$ and $w_* > 0$ a.e., this fails, so $\det G > 0$. ☐

The condition number $\kappa(\mathbf{G})$ quantifies identifiability (Fig. 1) (Fig. 1). For Maxwell–Boltzmann statistics ($\xi = 0$), κ is exactly η_0-independent: the Gram integrals factorise through $I_p^\xi(\eta_0) = \Gamma(p+1)\, e^{\eta_0}$ (Proposition 18), and the exponential factor cancels in the ratio. At $\Theta = 1.2$, $\kappa_{\mathrm{MB}} \approx 78$. For Fermi–Dirac, κ increases with η_0 due to Pauli blocking ($\kappa < 1000$ for $\eta_0 \lesssim 7$; monotonicity observed numerically at all tested configurations). For Bose–Einstein, κ remains below κ_{MB} for all $\eta_0 < 0$ (admissible domain): the Bose factor $(1+f)$ shifts Fisher weight toward lower x, reducing the correlation coefficient (equivalently increasing the principal angle) and lowering both the norm ratio and κ (Proposition 10(c)).

Figure 1 shows the eigenvalues, condition number, and principal angle as functions of η_0 for all three statistics. The BE admissible domain and its conditioning landscape are shown in Fig. 2.

C. Angular truncation and state variables

In the slab-symmetric case:

$$\begin{aligned} \Theta(\mu) &= \Theta_0\big(1 + A_\Theta\, \mu + Q_\Theta\, P_2(\mu)\big), \\ \eta(\mu) &= \eta_0 + A_\eta\, \mu\,, \end{aligned} \tag{6}$$

with gauge $\langle\Theta\rangle_\Omega = \Theta_0$. The state vector is $\boldsymbol{\alpha} = (\Theta_0, A_\Theta, Q_\Theta, \eta_0, A_\eta) \in \mathbb{R}^5$, and the observable vector is

$$\boldsymbol{m} = (\mathcal{E}_0, \mathcal{E}_1, \mathcal{E}_2, \mathcal{N}_0, \mathcal{N}_1)\,, \tag{7}$$

with $\mathcal{E}_\ell = \int P_\ell(\mu)\, \rho_E(\mu)\, \mathrm{d}\mu$ and $\mathcal{N}_\ell = \int P_\ell(\mu)\, \rho_N(\mu)\, \mathrm{d}\mu$, where $\rho_E = \int x^3 f\, \mathrm{d}x$ and $\rho_N = \int x^2 f\, \mathrm{d}x$. The choice of five observables is the minimal exactly-determined set.

For Bose–Einstein statistics, admissibility requires $\eta(\mu) \le 0$ for all $\mu \in [-1,1]$, i.e.,

$$\eta_0 + |A_\eta| \le 0\,. \tag{8}$$

The full admissible set is $\{(\eta_0, A_\eta) : \eta_0 < 0,\ |A_\eta| \le -\eta_0\}$. Without loss of generality one restricts to $A_\eta \ge 0$ by reversing the dipole axis $\mu \mapsto -\mu$; Fig. 2 displays this half-domain. Within this domain, $\kappa(G) < 200$ over all sampled configurations, so no additional safe-scope restriction is required.

D. The moment map and its Jacobian

Invertibility requires the 5×5 Jacobian $J_{i\alpha} = \partial m_i/\partial\alpha_\alpha$ to be full-rank. We evaluate J by central finite differences at nine configurations (three per statistics; Table II). In all cases $\mathrm{rank}(J) = 5$; the smallest singular value exceeds 0.011, and the condition number ranges from 233 to 3108. The $\partial/\partial A_\eta$ column projection residual onto the span of the other four columns is 0.5%–4.4% of the column norm: A_η is not a linear combination of the other parameters.

The result is stable under step-size variation from $\varepsilon = 10^{-3}$ to 10^{-8}: singular values agree to six significant digits. The observed rank-5 pattern persists across all nine representative parameter points for all three quantum statistics, indicating that non-redundancy is structural rather than a differencing artefact.

E. Indispensability of each observable

The five-observable set (7) is not merely sufficient; it is minimal in the sense that every observable is indispensable for full parameter identification.

Theorem 3 (Observable indispensability, all ξ)**.** *In the $(L_\Theta = 2, L_\eta = 1)$ truncation, neither the energy sector nor the number sector alone suffices to identify all five parameters. Specifically:*

(a) The energy sector alone $(\mathcal{E}_0, \mathcal{E}_1, \mathcal{E}_2)$ cannot identify either η_0 or A_η: the null space of the 3×5 sub-Jacobian contains the analytic null vector $n = (-3I_2^\xi\Theta_0/(4I_3^\xi), 0, 0, 1, 0)$.

(b) The number sector alone $(\mathcal{N}_0, \mathcal{N}_1)$ has rank $2 < 3$ and cannot close the temperature sector $(\Theta_0, A_\Theta, Q_\Theta)$.

Single-observable drops are characterised numerically in Proposition 4.

Proof. (a) By Theorem 14, $\mathcal{E}_k = I_3^\xi(\eta_0)\langle\mu^k\Theta^4\rangle$ for $k = 0,1,2$. Differentiating: $\partial\mathcal{E}_k/\partial\Theta_0 = (4/\Theta_0)\mathcal{E}_k$ and $\partial\mathcal{E}_k/\partial\eta_0 = (3I_2^\xi/I_3^\xi)\mathcal{E}_k$. The Θ_0- and η_0-columns are exactly proportional, creating the stated null vector. (b) Two observables can determine at most two parameter combinations. ☐

Proposition 4 (Numerical single-drop characterisation)**.** *At three representative configurations (one per statistics, Table* III*), the remaining single-observable drops are verified numerically via SVD: dropping $\mathcal{N}_1$ alone makes A_η the dominant unidentifiable direction (null-space alignment $\ge$ 93% with the (η_0, A_η) plane); dropping $\mathcal{N}_0$ alone makes η_0 the dominant unidentifiable direction. These results are stable under step-size variation across six orders of magnitude. This is a numerical characterisation over tested configurations.*

F. Analytic scalar-η no-go

Theorem 5 (Local scalar-η no-go on the pure-dipole temperature subfamily, all ξ)**.** *Let $\Theta(\mu) = \Theta_0(1 + A_\Theta\,\mu)$ with $Q_\Theta = 0$ and $0 < A_\Theta < 1$. Let $\mathcal{M}_{\mathrm{scalar}}$ denote the* *four-parameter subfamily with $A_\eta = 0$ and $\eta = \eta_0$ scalar. Let $\mathcal{M}_{\rm dipole}$ denote the full five-parameter manifold. For any admissible η_0, the observable $f_\mathcal{N}/f_\mathcal{E}$ takes the value $r_0 = \langle\mu\,\Theta^3\rangle/\langle\Theta^3\rangle/(\langle\mu\,\Theta^4\rangle/\langle\Theta^4\rangle)$ on the entire subfamily $\mathcal{M}_{\rm scalar}$, but shifts to $r_0 + C_\xi\,A_\eta + O(A_\eta^2)$ with $C_\xi > 0$ on $\mathcal{M}_{\rm dipole}$. Therefore no member of $\mathcal{M}_{\rm scalar}$ within the pure-dipole temperature subfamily ($Q_\Theta = 0$) can reproduce the observable image of $\mathcal{M}_{\rm dipole}$ at $A_\eta \neq 0$, even after arbitrary adjustment of $(\Theta_0, A_\Theta, \eta_0)$. Extension to the full manifold with $Q_\Theta \neq 0$ is supported numerically (Table VI, Remark 7).*

Proof. By Proposition 19, $f_\mathcal{N}/f_\mathcal{E} = r_0$ is constant on $\mathcal{M}_{\rm scalar}$. It remains to show $C_\xi > 0$. Using the derivative identity (Proposition 16), a first-order expansion in A_η gives

$$C_\xi = \frac{I_2^\xi/I_3^\xi}{f_\mathcal{E}}\left[2\,\frac{I_1^\xi I_3^\xi}{(I_2^\xi)^2}\,\frac{V_3}{V_4} - 3\,r_0\right]V_4\,, \tag{9}$$

where $V_p = \mathrm{Var}_{\Theta^p}(\mu) > 0$ is the Θ^p-weighted angular variance. All prefactors are strictly positive for $A_\Theta \neq 0$. The bracket is positive iff $I_1^\xi I_3^\xi/(I_2^\xi)^2 > (3/2)\,r_0\,(V_4/V_3)$. Two analytic inequalities close this chain.

Spectral inequality. By the Cauchy–Schwarz inequality, the map $p \mapsto \ln I_p^\xi$ is convex, giving $I_1^\xi I_3^\xi \geq (I_2^\xi)^2$. The infimum of $I_1^\xi I_3^\xi/(I_2^\xi)^2$ is 9/8, approached in the Sommerfeld limit $\eta_0 \to \infty$ (FD). The Sommerfeld expansion gives

$$\left.\frac{I_1^\xi I_3^\xi}{(I_2^\xi)^2}\right|_{\xi=-1} = \frac{9}{8}\left(1 + \frac{\pi^2}{3\eta_0^2} + O(\eta_0^{-4})\right) > \frac{9}{8}\,. \tag{10}$$

For MB: $I_1^\xi I_3^\xi/(I_2^\xi)^2 = 3/2$. For BE: log-convexity of the polylogarithm gives $I_1^\xi I_3^\xi/(I_2^\xi)^2 \geq 3/2 > 9/8$.

Angular inequality. Define $g(A_\Theta) := r_0\,(V_4/V_3)$. At $A_\Theta = 0$: $r_0 \to 3/4$ and $V_3/V_4 \to 1$, so $g(0) = 3/4$. For $A_\Theta > 0$, exponential-family mean monotonicity gives $\langle\mu\rangle_p$ strictly increasing in p and V_p strictly decreasing in p (the family $\Theta^p \propto \exp(p\ln\Theta)$ has $\ln\Theta(\mu)$ as the natural parameter; the L^p mean $\langle\mu\rangle_p$ is the expected value of μ under the tilted distribution, which is strictly monotone in the tilting parameter p when Θ is nonconstant). Hence $r_0 < 1$ and $V_4/V_3 < 1$, giving $g(A_\Theta) < 3/4$.

Combining. $I_1^\xi I_3^\xi/(I_2^\xi)^2 > 9/8 \geq (3/2)\,r_0\,(V_4/V_3)$, hence $C_\xi > 0$. □

Remark 6 (Axis-flip symmetry). For $A_\Theta < 0$, the result follows by the axis-reversal symmetry $\mu \mapsto -\mu$, which maps $(A_\Theta, A_\eta) \mapsto (-A_\Theta, -A_\eta)$ while preserving all angular-integrated observables. Thus Theorem 5 holds for all $A_\Theta \neq 0$.

Remark 7 (Scope of the analytic no-go). Theorem 5 is proved on the pure-dipole temperature subfamily ($Q_\Theta = 0$). The global numerical insufficiency (Table VI) confirms that the separation persists at finite A_η and at nonzero Q_Θ, but extension to the full five-parameter manifold with $Q_\Theta \neq 0$ is not claimed as an analytic theorem.

Remark 8 (Strength of the no-go). Theorem 5 is a first-order local impossibility: it uses only the first derivative C_ξ at $A_\eta = 0$. The global numerical insufficiency (Table VI) confirms that the separation persists at finite A_η. Together they establish that scalar η is insufficient both locally (analytic) and globally (numerical).

G. Asymptotic structure of $C_\xi(\eta_0)$

The response coefficient $C_\xi(\eta_0)$ is governed by the ratio of spectral integrals. Three asymptotic regimes are analytically accessible.

Maxwell–Boltzmann: exact constant. By Proposition 18, $C_{\rm MB}$ is η_0-independent. At $A_\Theta = 0.2$, $C_{\rm MB} \approx 0.333$.

Fermi–Dirac: large-degeneracy decay. For large η_0, the Sommerfeld expansion gives

$$\left.\frac{I_2^\xi}{I_3^\xi}\right|_{\xi=-1} = \frac{4}{3\eta_0}\left(\frac{1+\pi^2/\eta_0^2}{1+2\pi^2/\eta_0^2}\right)^{-1} + O(\eta_0^{-5})\,, \tag{11}$$

which tracks the computed ratio to $< 1\%$ for $\eta_0 \geq 5$. Since $C_{\rm FD}$ is proportional to this ratio, $C_{\rm FD} = O(1/\eta_0)$ as $\eta_0 \to \infty$.

Bose–Einstein: edge enhancement. As $\eta_0 \to 0^-$, $I_p^\xi \to \zeta(p{+}1)\,\Gamma(p{+}1)$. At the edge, $C_{\rm BE} \approx 0.65$—nearly twice the classical value. In the deep bulk ($\eta_0 \to -\infty$), $C_{\rm BE} \to C_{\rm MB}$; the convergence is exponential: $C_{\rm BE} - C_{\rm MB} = O(e^{\eta_0})$.

H. Conditioning asymptotics

The following spectral lemma underlies both the conditioning analysis and the response-coefficient monotonicity of §III F.

Lemma 9 (Log-concavity of normalised FD moments). Define $a_k := I_k^\xi/k!$ where $I_k^\xi(\eta_0) = \int_0^\infty x^k\,w_\xi(x)\,\mathrm{d}x$ and $w_\xi = f(1{+}\xi f)$ is the Fisher weight. For Fermi–Dirac ($\xi = -1$), the sequence $\{a_k\}_{k\geq 0}$ is strictly log-concave:

$$a_k^2 > a_{k-1}\,a_{k+1} \qquad (k \geq 1). \tag{12}$$

For Maxwell–Boltzmann, equality holds for all k.

Proof. The FD Fisher weight $w_{-1}(x) = e^{x-\eta}/(e^{x-\eta}+1)^2 = \frac{1}{4}\mathrm{sech}^2\big((x{-}\eta)/2\big)$ is strictly log-concave: $(\mathrm{d}^2/\mathrm{d}x^2)\log w = -\frac{1}{2}\mathrm{sech}^2\big((x{-}\eta)/2\big) < 0$. A positive log-concave function is a Pólya frequency function of order 2 (PF_2). By Karlin's basic composition theorem [5], the Laplace transform $M(t) = \int_0^\infty e^{tx}w(x)\,\mathrm{d}x$ inherits PF_2, which implies that its Taylor coefficients $\{M^{(k)}(0)/k!\} = \{I_k^\xi/k!\}$ form a log-concave sequence. Strictness follows because the FD weight is strictly log-concave (not exponential); the exponential (MB) case yields exact equality. □

Proposition 10 (Conditioning structure, all ξ)**.**

(a) For MB ($\xi = 0$), $\kappa(G)$ is exactly η_0-independent: $\kappa_{\mathrm{MB}} \approx 78$ at $\Theta = 1.2$ for all η_0.

(b) For FD ($\xi = -1$): $\kappa(G) > \kappa_{\mathrm{MB}}$ at $\eta_0 = 0$ (by the same two-input argument as (c), with the Dirichlet eta prefactor $(49/45)$ overcoming the zeta ratio). The norm ratio $\rho := G_{11}/G_{22} = I_5^\xi/(\Theta^2 I_3^\xi)$ is strictly increasing in η_0 (Lemma 9: $a_3a_4 \geq a_2a_5$ gives $5I_4^\xi I_3^\xi \geq 3I_2^\xi I_5^\xi$, hence $\mathrm{d}\rho/\mathrm{d}\eta_0 \geq 0$). At large degeneracy, Sommerfeld concentration drives $\kappa \to \infty$.

(c) For BE ($\xi = +1$) and all admissible $\eta_0 \leq 0$, $\kappa(G) < \kappa_{\mathrm{MB}}$ whenever $\Theta < \sqrt{20}$ (equivalently $\rho_{\mathrm{MB}} > 1$). The inequality tightens as $\eta_0 \to 0^-$ and $\kappa_{\mathrm{BE}} \to \kappa_{\mathrm{MB}}^-$ as $\eta_0 \to -\infty$.

Proof. (a) For MB, all Gram integrals contain the common factor e^{η_0} which cancels in every ratio. (b) The endpoint inequality follows from the same (ρ, r^2) framework as (c): at $\eta_0 = 0$, both $\rho_{\mathrm{FD}} > \rho_{\mathrm{MB}}$ and $r_{\mathrm{FD}}^2 > r_{\mathrm{MB}}^2$ (the Dirichlet eta prefactors $(1-2^{1-k})$ give $(49/45) \cdot \zeta(4)^2/[\zeta(3)\zeta(5)] > 1$). The ρ-monotonicity uses Lemma 9: $\mathrm{d}(I_5^\xi/I_3^\xi)/\mathrm{d}\eta_0 = (5I_4^\xi I_3^\xi - 3I_2^\xi I_5^\xi)/I_3^{\xi\,2}$, and $5I_4^\xi I_3^\xi \geq 3I_2^\xi I_5^\xi$ is equivalent to $a_3a_4 \geq a_2a_5$ (log-concavity at adjacent indices). At large η_0, the Sommerfeld expansion gives $\rho \sim \eta_0^2/5$ and $r^2 \to 1^-$, so $\kappa \to \infty$. (c) Three steps (unchanged from the previous proof). *Step 1 (correlation).* $r^2 = \mu_4^2/(\mu_3\mu_5)$. For MB, $r_{\mathrm{MB}}^2 = 4/5$. For BE, $r_{\mathrm{BE}}^2 = (4/5)\,\mathrm{Li}_4(x)^2/[\mathrm{Li}_3(x)\,\mathrm{Li}_5(x)] < 4/5$ by the Cauchy–Schwarz inequality on the polylogarithm series. *Step 2 (norm ratio).* $\rho_{\mathrm{BE}} = \rho_{\mathrm{MB}}\,\mathrm{Li}_5(x)/\mathrm{Li}_3(x) < \rho_{\mathrm{MB}}$ by term-by-term comparison. *Step 3 (κ-monotonicity).* $\partial\kappa/\partial\rho = 4(\rho-1)(1-r^2)/[D(\rho+1-D)^2] > 0$ and $\partial\kappa/\partial r^2 > 0$ for $\rho > 1$. Both inputs decrease from MB to BE, giving $\kappa_{\mathrm{BE}} < \kappa_{\mathrm{MB}}$. □

Remark 11 (FD κ-monotonicity: analytical and numerical status). Proposition 10(b) establishes three analytical results: the endpoint inequality $\kappa_{\mathrm{FD}}(0) > \kappa_{\mathrm{MB}}$, the monotone growth of ρ with η_0 (Lemma 9), and the Sommerfeld divergence $\kappa \to \infty$. Full monotonicity of $\kappa(\eta_0)$ additionally requires monotone growth of r^2, whose derivative involves a three-term combination of spectral ratios not directly controlled by one-step log-concavity. Numerically, κ increases monotonically across all tested configurations (Table IV: from 92 at $\eta_0 = 0$ to > 3000 at $\eta_0 = 10$).

I. Cross-sector observable family

Proposition 12 (Sector-separability, all ξ)**.** *Let $O_{p,q} := \langle \mu^p\Theta^q\rangle/\langle\Theta^q\rangle$ denote a within-sector angular moment ratio of order p with energy weight q. At scalar $\eta = \eta_0$ (i.e. $A_\eta = 0$), $O_{p,q}$ is independent of η_0 for all p and all q within a single sector. In contrast, any cross-sector ratio mixing different energy weights $q_1 \neq q_2$ acquires an A_η-dependent shift at first order whenever the spectral integrals $I_{q_1}^\xi$ and $I_{q_2}^\xi$ have different η_0-derivatives.*

Proof. By Theorem 14, $\langle \mu^p\Theta^q\rangle = I_q^\xi(\eta_0) \cdot \langle \mu^p\Theta^{q-1}\rangle_{\mathrm{ang}}$. The I_q^ξ factor cancels in same-sector ratios. Cross-sector ratios involve $I_{q_1}^\xi/I_{q_2}^\xi$, with nontrivial η_0-derivative by Proposition 16. □

Remark 13. For MB, even cross-sector ratios are η_0-invariant because $\mathrm{d}\ln I_p^\xi/\mathrm{d}\eta_0 = 1$ for all p (Remark 17). The chemical-potential dipole A_η still shifts cross-sector ratios, but the shift is η_0-independent.

The ratio $f_{\mathcal{N}}/f_{\mathcal{E}} = O_{1,3}/O_{1,4}$ is the simplest representative of the cross-sector family.

III. NON-REDUNDANCY AND OBSERVABLE SIGNATURE

A. Universal factorisation at scalar η

A structural identity underlies both the non-redundancy argument and the observable signature. When the chemical potential is a scalar ($\eta(\hat{e}) = \eta_0$, no angular dependence), the energy integration and the angular integration separate completely: the spectral shape contributes a direction-independent scalar prefactor, while all angular structure is encoded in powers of $\Theta(\hat{e})$. This separation is the algebraic origin of the scalar-η invariance of moment ratios (Proposition 19) and the starting point for the non-redundancy proof (Theorem 5).

Theorem 14 (Universal factorisation, all ξ)**.** *For any statistics $\xi \in \{0, -1, +1\}$, any positive angular temperature field $\Theta(\hat{e}) > 0$, scalar η_0, and angular kernel $\phi(\hat{e})$, the spectral moment of order p factorises:*

$$M_p[\phi] := \int \mathrm{d}\Omega\, \phi(\hat{e}) \int_0^\infty x^p\, \Phi_\xi\!\left(\frac{x}{\Theta(\hat{e})} - \eta_0\right) \mathrm{d}x = I_p^\xi(\eta_0) \int \mathrm{d}\Omega\, \phi(\hat{e})\, \Theta(\hat{e})^{p+1}, \quad (13)$$

where $I_p^\xi(\eta_0) := \int_0^\infty u^p/(\mathrm{e}^{u-\eta_0} - \xi)\,\mathrm{d}u$ is the generalised spectral integral. Consequently, any same-p normalised angular ratio is independent of η_0.

Proof. The ray-wise energy density with weight x^p is $\int_0^\infty x^p\,\Phi_\xi(x/\Theta - \eta_0)\,\mathrm{d}x$. Substituting $u = x/\Theta$ (so $\mathrm{d}x = \Theta\,\mathrm{d}u$ and $x^p = \Theta^p u^p$) gives $\Theta^{p+1}\int_0^\infty u^p\,\Phi_\xi(u - \eta_0)\,\mathrm{d}u = \Theta^{p+1} I_p^\xi(\eta_0)$. The substitution requires only $\Theta > 0$ and convergence of the integral, both of which hold for all admissible (Θ, η_0, ξ). The spectral integral I_p^ξ is independent of $\hat{e}$ and factors out of the angular integral. □

Remark 15. Theorem 14 holds for arbitrary $\Theta(\hat{e})$ and arbitrary angular kernel $\phi(\hat{e})$, not just the $(L_\Theta = 2)$ truncation. Proposition 19 is its immediate corollary.

Physical content. Two structural consequences follow. First, the direction dependence of the ray-wise densities enters only through powers of $\Theta(\mu)$; all η_0-dependence is sequestered into the spectral integrals, which are direction-independent scalars. Second, any angular observable constructed as a ratio of ρ-weighted integrals has the I_p^ξ factors cancel, rendering the observable η_0-independent. The energy sector uses $p = 3$ ($\rho_{\mathcal{E}} = \Theta^4 I_3^\xi$), the number sector uses $p = 2$ ($\rho_{\mathcal{N}} = \Theta^3 I_2^\xi$), and cross-sector ratios involve I_2^ξ/I_3^ξ, which *does* depend on η_0 for FD and BE (but not for MB).

The spectral integrals take distinct forms for each statistics:

$$I_p^\xi\big|_{\xi=0} = \Gamma(p+1)\,\mathrm{e}^{\eta_0} \quad \text{(MB)}, \tag{14}$$

$$I_p^\xi\big|_{\xi=-1} = F_p(\eta_0) \quad \text{(Fermi integral)}, \tag{15}$$

$$I_p^\xi\big|_{\xi=+1} = B_p(\eta_0) \quad \text{(Bose integral, } \eta_0 \le 0\text{)}. \tag{16}$$

The MB case is distinguished by *multiplicative separability*: I_p^ξ factors into an η_0-dependent part (e^{η_0}) and a p-dependent part ($\Gamma(p+1)$). Neither the Fermi integral F_p nor the Bose integral B_p admits such a factorisation—this asymmetry is the algebraic origin of the statistics-dependent response coefficient $C_\xi(\eta_0)$.

Worked example. At $\Theta(\mu) = 1 + 0.2\,\mu$ (dipole with $A_\Theta = 0.2$) and $\eta_0 = 3$ (moderate FD degeneracy): $I_3^\xi(\eta_0{=}3)\big|_{\xi=-1} = F_3(3) \approx 24.9$, $I_2^\xi(\eta_0{=}3)\big|_{\xi=-1} = F_2(3) \approx 6.23$. The ray-wise energy density at $\mu = +1$ is $\rho_{\mathcal{E}}(+1) = (1.2)^4 \times 24.9 \approx 51.6$; at $\mu = -1$: $(0.8)^4 \times 24.9 \approx 10.2$. The ratio $\rho_{\mathcal{E}}(+1)/\rho_{\mathcal{E}}(-1) = (1.2/0.8)^4 = 5.06$ is independent of η_0—the spectral integral I_3^ξ cancels.

B. Derivative identities

The response of the spectral integrals to changes in η_0 governs the η_0-dependence of C_ξ.

Proposition 16 (Spectral integral derivative, all ξ)**.** *For all admissible η_0 and $p \ge 1$,*

$$\frac{\mathrm{d}I_p^\xi}{\mathrm{d}\eta_0} = p\,I_{p-1}^\xi(\eta_0). \tag{17}$$

Proof. Differentiating under the integral sign:

$$\frac{\mathrm{d}I_p^\xi}{\mathrm{d}\eta_0} = \int_0^\infty \frac{u^p\,\mathrm{e}^{u-\eta_0}}{(\mathrm{e}^{u-\eta_0}-\xi)^2}\,\mathrm{d}u. \tag{18}$$

Integration by parts with the identity $\mathrm{e}^{u-\eta_0}/(\mathrm{e}^{u-\eta_0} - \xi)^2 = -(\mathrm{d}/\mathrm{d}u)[1/(\mathrm{e}^{u-\eta_0} - \xi)]$:

$$= \Big[-\frac{u^p}{\mathrm{e}^{u-\eta_0}-\xi}\Big]_0^\infty + p\int_0^\infty \frac{u^{p-1}}{\mathrm{e}^{u-\eta_0}-\xi}\,\mathrm{d}u. \tag{19}$$

The boundary term vanishes: at $u \to \infty$ the exponential dominates; at $u = 0$ the factor u^p vanishes for $p \ge 1$. The surviving integral is $p\,I_{p-1}^\xi$. □

Remark 17 (MB self-reproduction). For $\xi = 0$: $\mathrm{d}I_p^\xi/\mathrm{d}\eta_0 = \Gamma(p+1)\,\mathrm{e}^{\eta_0} = I_p^\xi$. The derivative reproduces the function—this follows algebraically from $p\,\Gamma(p) = \Gamma(p+1)$. Self-reproduction is unique to the exponential (MB) case and is the algebraic origin of the η_0-independence of C_{MB} (Proposition 18).

Physical interpretation. For quantum statistics, the derivative $\mathrm{d}I_p^\xi/\mathrm{d}\eta_0 = p\,I_{p-1}^\xi$ shifts the spectral weight from order p to order $p{-}1$: the perturbation preferentially samples lower-energy modes. Because the ratio I_{p-1}^ξ/I_p^ξ is not η_0-independent for FD or BE, the response coefficient C_ξ inherits an η_0-dependence. For FD at large degeneracy, $I_2^\xi/I_3^\xi \approx 4/(3\eta_0)$ (Sommerfeld expansion, Appendix A 3), so $C_{\mathrm{FD}} = O(1/\eta_0)$: the response decreases because Pauli blocking saturates progressively more energy modes.

C. MB η_0-independence

Proposition 18 (MB η_0-independence of C_ξ)**.** *For Maxwell–Boltzmann statistics, the first-order response coefficient $C_\xi(\eta_0)$ is independent of η_0.*

Proof. At $A_\eta = 0$, the flux fractions are $f_{\mathcal{E}} = \langle\mu\Theta^4\rangle/\langle\Theta^4\rangle$ and $f_{\mathcal{N}} = \langle\mu\Theta^3\rangle/\langle\Theta^3\rangle$, both independent of η_0 by Theorem 14. The first-order perturbation in A_η introduces the factor $\mu\,\mathrm{d}I_p^\xi/\mathrm{d}\eta_0$ into the moment integrals. By Remark 17, $\mathrm{d}I_p^\xi/\mathrm{d}\eta_0 = I_p^\xi$ for $\xi = 0$, so the η_0-dependent factor e^{η_0} cancels in every ratio that defines C_ξ. □

This is a structurally remarkable property: for classical (Boltzmann) statistics, the sensitivity of the number-to-energy flux ratio to chemical-potential anisotropy depends only on the angular geometry of $\Theta(\hat{e})$, not on the overall degeneracy level. For quantum statistics, the degeneracy level modulates the sensitivity through the ratio I_2^ξ/I_3^ξ.

D. Cross-sector observable family

The flux fraction ratio $f_{\mathcal{N}}/f_{\mathcal{E}}$ is not an isolated example; it is the lowest-order member of the broader class of cross-sector ratios identified in Proposition 12 (§II I).

The ratio $f_{\mathcal{N}}/f_{\mathcal{E}} = O_{1,3}/O_{1,4}$ mixes $q = 3$ (number) and $q = 4$ (energy) with the lowest angular order $p = 1$. Higher-order cross-sector ratios (e.g. $O_{2,3}/O_{2,4}$) carry the same structural sensitivity but are harder to measure observationally.

E. Scalar-η insufficiency

The analytic no-go theorem (Theorem 5) establishes that the scalar subfamily cannot reproduce the dipolar observable image at first order. We complement this with a global numerical test: fix $\Theta(\mu) = 1 + A_\Theta\mu$ with $A_\Theta =$

0.2, set $\eta(\mu) = \eta_0 + A_\eta \mu$ with $A_\eta = 0.5$, and attempt to match the resulting five moments using the scalar model $\eta = \eta_0^*$ with free parameters $(\Theta_0, A_\Theta, Q_\Theta, \eta_0^*)$. The normalised moment residual after full Nelder–Mead reoptimisation is

$$R_{\mathrm{scalar}} = \min_{\Theta_0, A_\Theta, Q_\Theta, \eta_0^*} \frac{\|\boldsymbol{m}[\eta{=}\eta_0{+}A_\eta\mu] - \boldsymbol{m}[\eta{=}\eta_0^*]\|}{\|\boldsymbol{m}[\eta{=}\eta_0{+}A_\eta\mu]\|}\,. \tag{20}$$

Table VI reports the residuals: they range from 5.2×10^{-4} (FD, $\eta_0 = 6$) to 1.2×10^{-2} (BE, $\eta_0 = -0.5$), all 6–10 orders of magnitude above the quadrature floor ($\sim 10^{-12}$).

Physical interpretation. The residuals are largest near the BE condensation edge ($\eta_0 = -0.5$), where Bose enhancement amplifies the number-sector response, and smallest at deep FD degeneracy ($\eta_0 = 6$), where Pauli blocking compresses the distribution toward uniformity. Crucially, the residual never reaches the quadrature floor at any degeneracy: the information carried by $L_\eta = 1$ is structurally present in the moment vector for all ξ and all η_0.

F. Observable signature: the flux fraction ratio

Proposition 19 (Scalar-η invariance of $f_\mathcal{N}/f_\mathcal{E}$, all ξ)**.** *At $A_\eta = 0$, the flux fraction ratio $f_\mathcal{N}/f_\mathcal{E}$ is independent of η_0 for all three quantum statistics.*

Proof. At constant $\eta = \eta_0$, Theorem 14 gives

$$f_\mathcal{E} = \frac{\int \mu\, \Theta^4\, \mathrm{d}\mu}{\int \Theta^4\, \mathrm{d}\mu}, \qquad f_\mathcal{N} = \frac{\int \mu\, \Theta^3\, \mathrm{d}\mu}{\int \Theta^3\, \mathrm{d}\mu}. \tag{21}$$

The $I_p^\xi(\eta_0)$ factors cancel in both numerator and denominator, and hence in the ratio. □

Numerical verification. At $A_\Theta = 0.2$, the flux fraction ratio $f_\mathcal{N}/f_\mathcal{E} = 0.766904$ is η_0-independent to 12–15 significant digits across all tested η_0 values for all three statistics (Table V). The residual spread ($\sim 10^{-11}$) is set by the angular quadrature precision, not by any physical η_0-dependence. This exact invariance is the cleanest observable consequence of the factorisation theorem.

The η-dipole breaks this invariance: when $\eta(\mu) = \eta_0 + A_\eta \mu$, the spectral integrals acquire direction dependence (through the $\eta(\mu)$-dependent argument) and the cancellation fails. The first-order response coefficient is:

Proposition 20 (First-order response coefficient, all ξ)**.** *The coefficient*

$$C_\xi(\eta_0) := \left.\frac{\mathrm{d}}{\mathrm{d}A_\eta} \frac{f_\mathcal{N}}{f_\mathcal{E}}\right|_{A_\eta=0} \tag{22}$$

satisfies: (a) *MB: C_0 is η_0-independent (Proposition 18).* (b) *FD: the spectral ratio I_2^ξ/I_3^ξ that governs C_{-1} is strictly decreasing in η_0 (Lemma 9: $a_2^2 \geq a_1 a_3$ gives $3(I_2^\xi)^2 \geq 2I_1^\xi I_3^\xi$, hence $\mathrm{d}(I_2^\xi/I_3^\xi)/\mathrm{d}\eta_0 \leq 0$). At large degeneracy, $C_{-1} = O(1/\eta_0)$ (Sommerfeld expansion).* (c) *BE: $C_{+1} \to C_{\mathrm{MB}}$ in the non-degenerate bulk with exponential convergence $C_{+1} - C_{\mathrm{MB}} = O(\mathrm{e}^{\eta_0})$. At the condensation edge ($\eta_0 \to 0^-$), the spectral ratio reaches the exact value $I_2^\xi/I_3^\xi \to \pi^2/[18\zeta(3)] \approx 0.456$, exceeding the MB value $1/3$ by a factor $\pi^2/[6\zeta(3)] \approx 1.37$.*

Remark 21 (Full C_ξ monotonicity). The full coefficient (22) involves the spectral ratio I_2^ξ/I_3^ξ (proved monotone decreasing for FD by Lemma 9) together with a second spectral factor $I_1^\xi I_3^\xi/(I_2^\xi)^2$ that also decreases numerically. Since the angular factors (V_3, V_4, r_0, f_E) are η_0-independent at scalar η_0 (Theorem 14), and both spectral factors move in the same direction, C_{-1} is observed to decrease monotonically with η_0 across all tested configurations (Table VII). The corresponding BE pattern—C_{+1} decreasing from the condensation edge ($C_{+1} \approx 0.65$ at $\eta_0 = -0.5$) toward the MB value—is likewise observed numerically (Table VII).

Physical origins (Table VII, Fig. 3). The MB value $C_{\mathrm{MB}} \approx 0.333$ is a pure angular-geometry functional of $\Theta(\mu)$; its η_0-independence reflects MB self-reproduction (Remark 17). The FD decrease is driven by Pauli blocking: as the Fermi level rises, the occupation $f \to 1$ over an increasing energy range, suppressing the sensitivity to η-perturbations. Quantitatively, the ratio I_2^ξ/I_3^ξ—which governs the coupling between the η-perturbation and the number-flux moment—decreases from 0.317 ($\eta_0 = 0$) to 0.179 ($\eta_0 = 6$), directly tracking the decline of C_{FD}. The BE enhancement near the condensation edge originates from the Bose factor $f(1{+}f)$, which amplifies the number-sector response (weight x^2) more strongly than the energy-sector response (weight x^3).

The flux fraction ratio $f_\mathcal{N}/f_\mathcal{E}$ is the cleanest diagnostic of the η-dipole: exactly invariant under scalar η_0 (Proposition 19), responsive at first order in A_η with a physically transparent coefficient, and accessible to any transport code tracking both energy and number moments.

G. Pure η-dipole observable loading

Proposition 22 (Pure η-dipole observable loading)**.** *At $A_\Theta = 0$, $A_\eta \neq 0$:*

(a) *The leading observable effect enters the number sector: $\mathcal{N}_1 = O(A_\eta)$ while $\chi - 1/3 = O(A_\eta^2)$.*

(b) *On the one-field Teff manifold, the tangency diagnostic misclassifies all η-anisotropy as off-manifold content. On the two-field manifold, the Pythagorean decomposition (Paper III, Proposition 5) partitions the 1D diagnostic exactly. At moderate FD degeneracy ($\eta_0 \approx 3$), the η-tangent component accounts for 98.5% of $[\mathcal{D}_{\geq 2}^{(\mathrm{1D})}]^2$ (Table X).*

Physical significance. The η-sector is a number-sector decoupling variable, not an energy-scale anisotropy. The pure η-dipole generates a number flux without generating an energy flux (at leading order), breaking the lockstep between energy and number transport that the scalar-η manifold enforces. This is precisely the physics of neutrino charged-current processes, where the angular distribution of lepton number can differ from the angular distribution of energy.

IV. KNOWN-LIMIT CONSISTENCY

The ($L_\Theta = 2$, $L_\eta = 1$) manifold recovers the correct behaviour in five limiting regimes (Table VIII). Each limit is tested for all three statistics; the verification confirms that the non-redundancy structure established in §§II–III is consistent with the physics of the known-limit endpoints.

L1. Non-degenerate limit ($\eta_0 \to -\infty$, $A_\eta = 0$). As $\eta_0 \to -\infty$, the FD and BE distributions converge to Maxwell–Boltzmann: the quantum factor $(1+\xi f)$ tends to unity when $f \ll 1$. At $\eta_0 = -12$, the flux factor and Eddington factor differences between FD/BE and MB fall below 10^{-15} (Table IX, rows L1). At $A_\eta \neq 0$, the η-dipole effect persists because the exponential factor $e^{A_\eta \mu} \neq$ const survives the classical limit: the mathematical independence of A_η is present at all degeneracies, though its observational impact is strongest at moderate degeneracy. For BE, $C_{\rm BE}$ converges to $C_{\rm MB} = 0.333$ to six significant digits at $\eta_0 = -8$ (Table VII).

L2. Isotropic limit ($A_\Theta = A_\eta = 0$). At vanishing anisotropy: homogeneous equilibrium. The flux factor $|f_\mathcal{E}| < 10^{-14}$, the Eddington factor $|\chi - 1/3| < 10^{-15}$, and the monopole moments match analytic spectral integrals (14)–(16) to $< 10^{-10}$ relative error for all three statistics (Table IX, rows L2). These values confirm that the two-field manifold reduces to the correct isotropic equilibrium when all anisotropy parameters vanish.

L3. Scalar-η reduction ($A_\eta = 0$). At vanishing chemical-potential dipole, the manifold reduces to the scalar-chemical-potential extension. The flux fraction ratio $f_\mathcal{N}/f_\mathcal{E} = 0.766904$ is η_0-independent to 12 significant digits for all three statistics (Table IX, rows L3), confirming Proposition 19. The residual spread ($\sim 10^{-11}$) is set by the angular quadrature precision, not by any physical η_0-dependence. This is the numerical face of the factorisation theorem (Theorem 14): the spectral integrals cancel exactly in the flux-fraction ratio.

L4. Pure η-dipole ($A_\Theta = 0$, $A_\eta \neq 0$). Anisotropy from direction-dependent degeneracy alone. The linear prediction based on C_ξ (22) matches the numerical flux fraction ratio to $< 1\%$ for $A_\eta \leq 0.05$ across all statistics. The Eddington factor perturbation $\chi - 1/3$ is second-order in A_η, confirming that the leading observable effect is in the number sector (Proposition 22(a)). This limit tests the physical content of the non-redundancy claim: the η-dipole generates a number flux without generating an energy flux (at leading order), demonstrating that the number and energy sectors are genuinely decoupled.

L5. Conditioning characterisation. For FD, $\kappa(\mathbf{G})$ exceeds 1000 at $\eta_0 \approx 7$ (Table I). For BE, $\kappa(\mathbf{G}) < \kappa_{\rm MB}$ throughout the admissible domain $\eta_0 \leq 0$—a consequence of Bose enhancement (Proposition 10(c)). For MB, $\kappa = 78$ is η_0-independent (Proposition 10(a)). The Gram matrix and Jacobian remain well-defined in all cases; high conditioning is a practical caveat for numerical inversion, not a structural obstruction to the non-redundancy result.

V. DISCUSSION

A. What this paper establishes

The central result is that, within the ($L_\Theta = 2, L_\eta = 1$) effective-temperature manifold, the chemical-potential dipole A_η is a non-redundant nonequilibrium degree of freedom for all three quantum statistics. Four independent lines of evidence support this: (i) scalar-η insufficiency at the global moment level (Sec. III E); (ii) full-rank Jacobian at the local linear level (Sec. II D); (iii) a universal factorisation (Theorem 14) sequestering all η_0-dependence into direction-independent spectral integrals; and (iv) an exactly invariant observable $f_\mathcal{N}/f_\mathcal{E}$ (Proposition 19) whose first-order response is governed by a statistics-dependent coefficient C_ξ (Proposition 20).

B. What this paper does not establish

Non-redundancy (the information is independent) and utility (the information improves a specific calculation) are distinct claims. We establish the former, not the latter. Theorem 5 is proved on the pure-dipole temperature subfamily ($Q_\Theta = 0$); global extension to the full manifold with $Q_\Theta \neq 0$ is a numerical characterisation, not an analytic theorem (Remark 7). We do not prove the entropy-projection inequality for the extended manifold, nor the symmetric hyperbolicity of the corresponding Godunov system.

C. Microphysical motivation

In core-collapse supernovae, neutrino transport involves both energy and lepton-number fluxes through an optically thick medium with steep radial gradients. The electron-type neutrino (ν_e) and its antiparticle ($\bar{\nu}_e$) carry opposite lepton number; their angular distributions develop direction-dependent chemical potentials through charged-current absorption and emission rates. A scalar fugacity cannot represent the resulting anisotropy in the number-flux sector independently of the energy sector. The two-field manifold is the minimal extension that decouples these sectors.

D. Bose–Einstein admissibility

The BE branch requires $\eta(\hat{e}) \leq 0$ for all directions (8). Unlike Fermi–Dirac, the BE branch is well-conditioned throughout the scanned admissible domain ($\kappa < 200$ at all tested configurations; Fig. 2). The physical origin is Bose enhancement: the weight $f(1+f)$ grows as $\eta \to 0^-$, amplifying both Gram eigenvalues and improving their separation.

E. Implications for the diagnose-to-switch protocol

Paper I [1] introduces a diagnose-to-switch protocol on the one-field manifold. On the two-field manifold the diagnostic changes in two ways.

False-trigger elimination. The Pythagorean decomposition (Paper III [4], Proposition 5) partitions the 1D diagnostic exactly:

$$|\mathcal{D}^{(1D)}_{\geq 2}|^2 = |\Pi_{V_2 \ominus 1} G_\xi|^2_* + |\mathcal{D}^{(2D)}_{\geq 2}|^2 \,. \tag{23}$$

The first term is the η-tangent component that the one-field manifold misclassifies as off-manifold; the second is the genuine spectral distortion. At moderate FD degeneracy ($\eta_0 \approx 3$), the η-tangent component accounts for 98.5% of $[\mathcal{D}^{(1D)}]^2$ (Table X; Fig. 1, right panel).

Fingerprint clarification. At FD $\eta_0 = 3$, the 1D coefficient is 20× larger than the 2D coefficient; the 1D fingerprint prescribes a one-mode correction for tangent-space content. The 2D fingerprint correctly identifies the residual as broadly distributed (13% leading-mode concentration), signalling multigroup discretisation.

F. Relation to scalar-fugacity closures

In the Romatschke–Strickland tradition [2, 3, 6], the chemical potential is a scalar λ and the anisotropy is parameterised in momentum space. In the present framework, $\eta(\hat{e})$ carries direction dependence, and $f_\mathcal{N}/f_\mathcal{E}$ can be adjusted independently. The two frameworks are complementary.

G. Generality of the structural results

Two results of this paper hold beyond the ($L_\Theta = 2$, $L_\eta = 1$) truncation. First, the x-coordinate well-posedness (Theorem 1) depends only on $\Theta > 0$ and the admissibility of η, not on the angular truncation order; it extends to arbitrary measurable $\Theta(\hat{e})$ and $\eta(\hat{e})$, provided the relevant energy-weighted angular moments remain finite. Second, the same-sector ratio invariance (Proposition 19) is a special case of a general factorisation property: any ratio of observables sharing the same energy weight p is η_0-independent at scalar η, because the spectral integral $\mathcal{I}^{(p)}_\xi(\eta_0)$ cancels in numerator and denominator. For MB, even cross-sector ratios are η_0-independent (Remark 13). The ($L_\Theta = 2$, $L_\eta = 1$) truncation is a practical choice, not a structural requirement.

TABLE I. Gram matrix eigenvalues and condition number at $\Theta = 1.2$ for representative η_0. MB: κ is η_0-independent (Proposition 18). FD: κ increases with η_0 (Pauli blocking). BE: $\kappa < \kappa_{\rm MB}$ throughout the admissible domain (8); no safe-scope threshold is needed.

ξ	η_0	$\lambda_{\min}$	$\lambda_{\max}$	$\kappa(\mathbf{G})$
0 (MB)	any	0.27	21	78
−1 (FD)	0	0.31	29	92
	3	0.99	3.5×10^2	355
	6	4.1	6.6×10^3	1604
+1 (BE)	−0.1	2.9	1.7×10^2	58
	−1	0.96	68	71
	−5	0.016	1.2	78

H. Outlook

Three developments build directly on the present work. First, Paper III [4] develops the information geometry of the two-field manifold. Second, the entropy-projection inequality of [1] should be extended to the (x, Θ, η) coordinate system. Third, coupling the five-parameter state vector to a transport solver with an explicit lepton-number equation would test whether the non-redundancy translates to a practical closure advantage.

Paper II establishes structural non-redundancy and identifiability of the η-dipole; it does not claim a transport-performance improvement relative to the one-field manifold. Any later operational statement that attributes cross-sector ratios solely to Θ-multipoles must be understood as restricted to the scalar-η subfamily; the present paper shows that anisotropic $\eta(\hat{e})$ changes that conclusion already at first order.

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Data availability. The Python code used to generate all numerical results and figures is available as Supplemental Material [URL will be inserted by publisher]. No experimental data were generated or analysed.

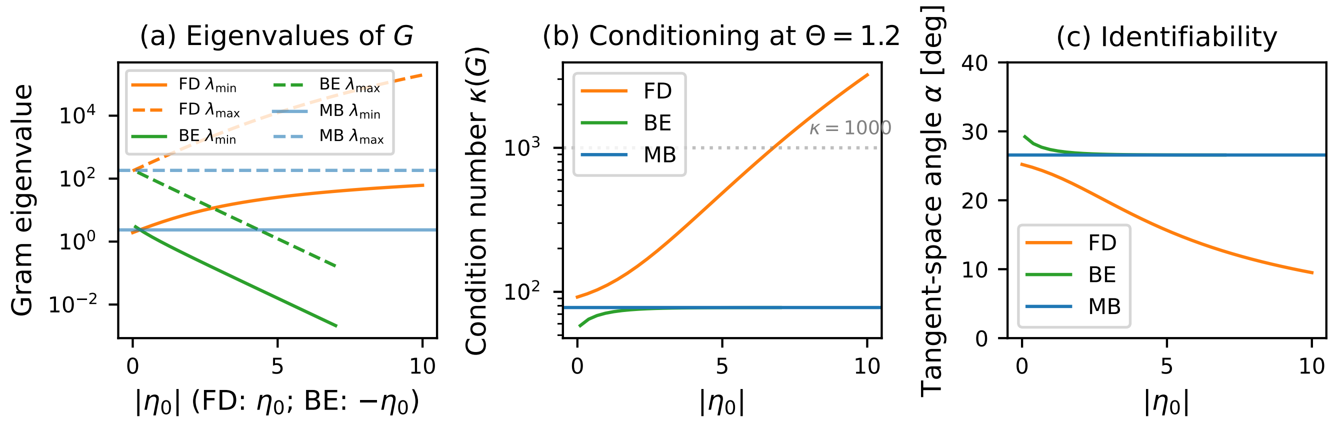

FIG. 1. Gram matrix eigenvalues, condition number, and tangent-space angle for the two-field manifold at $\Theta = 1.2$. (*a*) Eigenvalues $\lambda_{\min}$ (solid) and $\lambda_{\max}$ (dashed) versus η_0 for MB (blue), FD (orange), and BE (green). (*b*) Condition number $\kappa(\mathbf{G})$: MB is constant (≈ 78); FD increases monotonically; BE remains below 78 throughout. (*c*) Tangent-space angle α: all statistics maintain $\alpha > 10^\circ$, confirming linear independence.

TABLE II. 5×5 moment-map Jacobian at nine test configurations (three per statistics). All configurations have rank $= 5$; A_η column residual measures the projection of the $\partial/\partial A_\eta$ column onto the span of the other four.

ξ	Config	$\sigma_{\min}$	$\sigma_{\max}$	$\kappa(J)$	A_η resid. (%)
MB	1	0.14	32	234	4.4
	2	0.94	308	327	2.7
	3	0.25	60	238	3.7
FD	1	0.43	394	907	1.5
	2	0.60	1861	3108	0.5
	3	0.35	139	392	2.5
BE	1	0.058	14	233	4.1
	2	0.017	6.9	408	2.1
	3	0.011	4.6	425	1.9

TABLE III. Null-direction analysis: when each observable is dropped, the null vector of the 4×5 sub-Jacobian identifies the unrecoverable parameter direction. $\|\cdot\|_\eta$: projection onto the (η_0, A_η) plane. All three statistics show the same structural pattern.

Drop	Dominant null	$\|\cdot\|_\eta$	$\|\cdot\|_\Theta$
$\mathcal{E}_0$	η_0	0.93	0.37
$\mathcal{E}_1$	A_η	0.95	0.32
$\mathcal{E}_2$	Q_Θ (mixed)	0.53	0.85
$\mathcal{N}_0$	η_0	0.97	0.24
$\mathcal{N}_1$	A_η	0.97	0.24

TABLE IV. Conditioning structure of the Gram matrix at $\Theta = 1.2$. α: principal angle between tangent directions, $\cos\alpha = G_{12}/\sqrt{G_{11}G_{22}}$. η-align.: alignment of the soft eigenvector with the η-direction $|\langle e_{\rm soft}, \hat{e}_\eta\rangle|$. MB is exactly η_0-independent; FD degrades with degeneracy; BE improves near the edge.

ξ	η_0	$\kappa(\mathbf{G})$	α [°]	η-align.
0 (MB)	any	78	26.6	0.972
-1 (FD)	0	92	25.2	0.973
	3	212	19.7	0.980
	6	736	14.0	0.989
	10	3207	9.5	0.994
$+1$ (BE)	-0.3	63	28.5	0.970
	-1	71	27.3	0.971
	-3	77	26.7	0.972

Appendix A: Numerical verification details

1. Quadrature convergence

All moment integrals are evaluated by Gauss–Legendre quadrature with N_μ angular points and N_x energy points on the interval $[0, x_{\max}]$. The flux fraction ratio is converged to 12 significant digits at $N_\mu = 64$, $N_x = 128$, the resolution used throughout the paper.

2. Jacobian step-size stability

The moment-map Jacobian is computed by central finite differences with step size ε. Stability across six orders of magnitude (10^{-3} to 10^{-8}): smallest singular value, condition number, and A_η-column residual are stable to six significant digits.

TABLE V. Cross-sector vs. within-sector angular-moment ratios at $A_\Theta = 0.2$, $A_\eta = 0$. All ratios are η_0-independent to quadrature precision (Theorem 14). At $A_\eta \neq 0$: cross-sector ratios (mixing $q = 3$ and $q = 4$, e.g. $f_\mathcal{N}/f_\mathcal{E}$) acquire a statistics-dependent first-order shift, while same-sector ratios respond only through higher-order angular coupling.

ξ	η_0	$\chi_\mathcal{E}$	$\chi_\mathcal{N}$	$f_\mathcal{N}/f_\mathcal{E}$
MB	0	0.3532	0.3436	0.7669
	3	0.3532	0.3436	0.7669
	8	0.3532	0.3436	0.7669
FD	0	0.3532	0.3436	0.7669
	3	0.3532	0.3436	0.7669
	6	0.3532	0.3436	0.7669
BE	−0.5	0.3532	0.3436	0.7669
	−3	0.3532	0.3436	0.7669
	−8	0.3532	0.3436	0.7669

TABLE VI. Scalar-η insufficiency: normalised moment residual R_scalar Eq. (13) of the text after full reoptimisation of $(\Theta_0, A_\Theta, Q_\Theta, \eta_0)$, with $A_\eta = 0.5$, $A_\Theta = 0.2$. The quadrature floor is $\sim 10^{-12}$; all residuals exceed it by 6–10 orders of magnitude.

ξ	η_0	R_scalar
0 (MB)	0	8.1×10^{-3}
	3	8.1×10^{-3}
−1 (FD)	0	6.1×10^{-3}
	3	2.1×10^{-3}
	6	5.2×10^{-4}
+1 (BE)	−0.5	1.2×10^{-2}
	−2	8.5×10^{-3}
	−5	8.1×10^{-3}

3. Sommerfeld expansion for C_FD

The Fermi integral of order p admits the Sommerfeld expansion for large η_0:

$$I_p^\xi\big|_{\xi=-1} = \frac{\eta_0^{p+1}}{p+1}\left(1 + \frac{p(p+1)\pi^2}{6\eta_0^2} + O(\eta_0^{-4})\right). \tag{A1}$$

The ratio governing C_FD is

$$\frac{I_2^\xi}{I_3^\xi} = \frac{4}{3\eta_0}\cdot\frac{1+\pi^2/\eta_0^2}{1+2\pi^2/\eta_0^2} + O(\eta_0^{-5})\,. \tag{A2}$$

At $\eta_0 \geq 20$ the second-order approximation is accurate to $< 0.03\%$. Since C_FD is a smooth function of this ratio, the leading asymptotic $C_\text{FD} = O(1/\eta_0)$ follows.

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TABLE VII. First-order response coefficient $C_\xi(\eta_0)$ at $A_\Theta = 0.2$. The baseline ratio $f_\mathcal{N}/f_\mathcal{E}|_{A_\eta=0} = 0.7669$ is universal (Proposition 19). The ratio I_2^ξ/I_3^ξ quantifies the spectral-weight shift that governs C_ξ (Sec. III F).

ξ	η_0	C_ξ	I_2^ξ/I_3^ξ
0 (MB)	any	0.333	1/3
−1 (FD)	0	0.269	0.317
	1	0.218	0.289
	3	0.119	0.250
	6	0.048	0.179
+1 (BE)	−0.5	0.424	0.350
	−1	0.378	0.342
	−3	0.338	0.334
	−8	0.333	0.333

TABLE VIII. Known-limit consistency: summary. PASS indicates agreement to stated precision; CHAR indicates characterisation without a pass/fail threshold.

Limit	Property	ξ	Status
L1	Non-degenerate reduction	all	PASS
L2	Isotropic equilibrium	all	PASS
L3	Scalar-η invariance (Prop. 19)	all	PASS
L4	η-dipole linearity	all	PASS
L5	Conditioning	all	CHAR

TABLE IX. Known-limit numerical verification. Δf: flux factor deviation from reference; $\Delta\chi$: Eddington factor deviation; $\Delta\mathcal{E}_0$: relative monopole deviation from analytic spectral integral; σ_r: spread of $f_\mathcal{N}/f_\mathcal{E}$ across tested η_0 values.

	ξ	Quantity	Reference	Result
L1	FD	$\|\Delta f\|$	MB	$< 10^{-15}$
	BE	$\|\Delta f\|$	MB	$< 10^{-15}$
L2	MB	$\|f_\mathcal{E}\|$	0	$< 10^{-14}$
	FD	$\|\chi - 1/3\|$	0	$< 10^{-15}$
	BE	$\|\Delta\mathcal{E}_0\|$	I_3^ξ	$< 10^{-10}$
L3	MB	σ_r	0	5.6×10^{-16}
	FD	σ_r	0	1.8×10^{-11}
	BE	σ_r	0	9.1×10^{-12}

TABLE X. Upgrade decision comparison: 1D (one-field) vs 2D (two-field) diagnostic at $A_\Theta = 0.2$, $\mu = 0.8$. η-frac.: fraction of $[\mathcal{D}^{(1D)}_{\geq 2}]^2$ from η-tangent content (false trigger source). c_2 conc.: leading-mode concentration $|\kappa_2|^2/D^2$ (determines one-mode vs multigroup).

ξ	η_0	η-frac.	c_2 (1D)	c_2 (2D)	Prescription
MB	0	84%	94%	91%	one-mode
FD	0	89%	92%	89%	one-mode
	3	98.5%	81%	13%	multigroup†
	6	92%	68%	56%	multigroup
BE	-0.5	78%	95%	92%	one-mode
	-3	84%	94%	91%	one-mode

†1D diagnostic would prescribe one-mode (81% concentration), but 98.5% of its signal is η-tangent; the 2D diagnostic correctly identifies only 13% leading-mode concentration.

TABLE XI. Quadrature convergence of $f_\mathcal{N}/f_\mathcal{E}$ at $A_\Theta = 0.2$, $\eta_0 = 3$ (FD), $A_\eta = 0$. Reference value: $0.76690384616\ldots$

N_μ	N_x	$f_\mathcal{N}/f_\mathcal{E}$	Digits
16	32	0.766903846	9
32	64	0.7669038462	10
64	128	0.766903846163	12
128	256	0.766903846163	12

TABLE XII. Jacobian step-size stability (MB, config 1).

ε	$\sigma_{\min}$	$\kappa(J)$	A_η resid. (%)
10^{-3}	0.138617	233.6	4.44
10^{-4}	0.138616	233.6	4.44
10^{-5}	0.138616	233.6	4.44
10^{-6}	0.138616	233.6	4.44
10^{-7}	0.138616	233.6	4.44
10^{-8}	0.138616	233.6	4.44

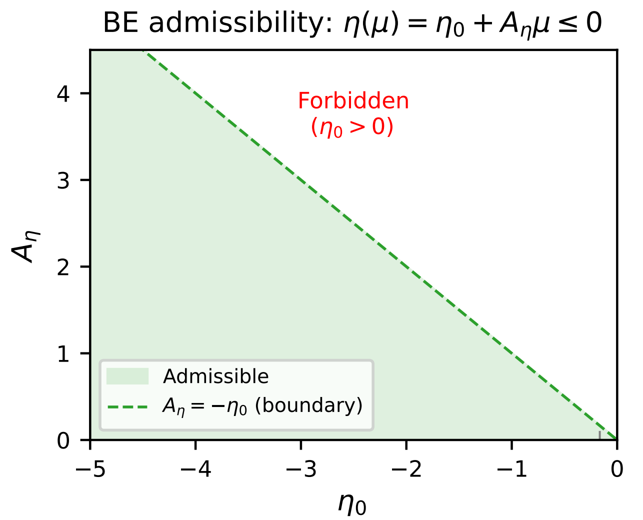

FIG. 2. BE admissibility domain. The shaded triangle $\{(\eta_0, A_\eta) : \eta_0 < 0,\ 0 \leq A_\eta \leq -\eta_0\}$ satisfies $\eta(\mu) \leq 0$ for all μ. The full admissible set includes $A_\eta < 0$ by axis reversal $\mu \mapsto -\mu$. Contours show the Gram condition number $\kappa(\mathbf{G})$; the maximum is ≈ 164, well below the FD safe-scope threshold of 1000.

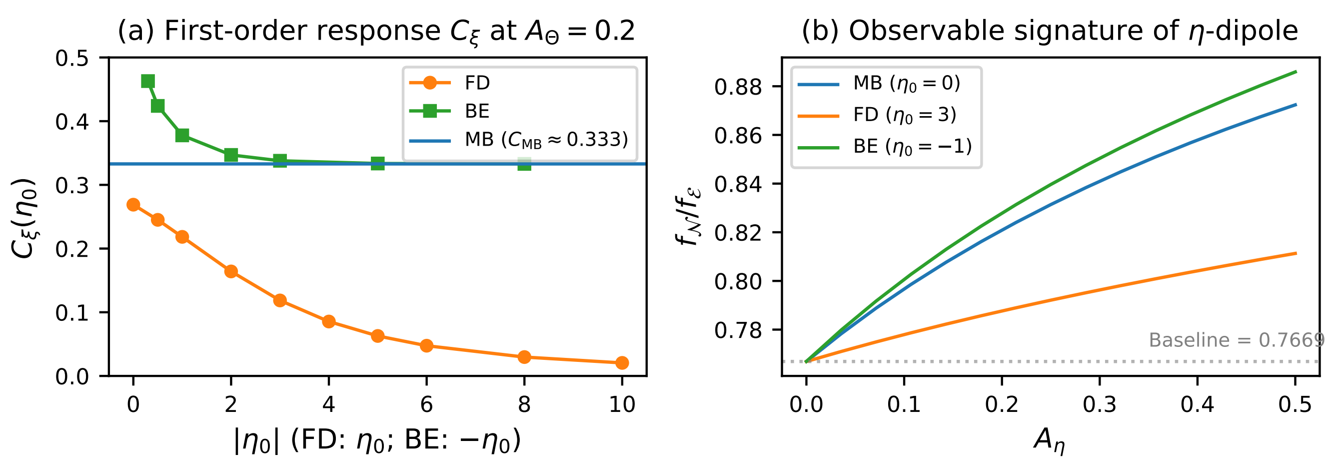

FIG. 3. (a) First-order response coefficient $C_\xi(\eta_0)$ at $A_\Theta = 0.2$: MB (blue) is η_0-independent (Proposition 18); FD (orange) decreases monotonically due to Pauli blocking, tracking the decline of I_2^ξ/I_3^ξ; BE (green) decreases from the condensation edge (Bose enhancement) and converges to $C_{\rm MB}$ in the non-degenerate limit. (b) Flux fraction ratio $f_\mathcal{N}/f_\mathcal{E}$ versus A_η: all statistics share the universal baseline 0.7669 at $A_\eta = 0$ (Proposition 19); the steepness of departure reflects C_ξ.